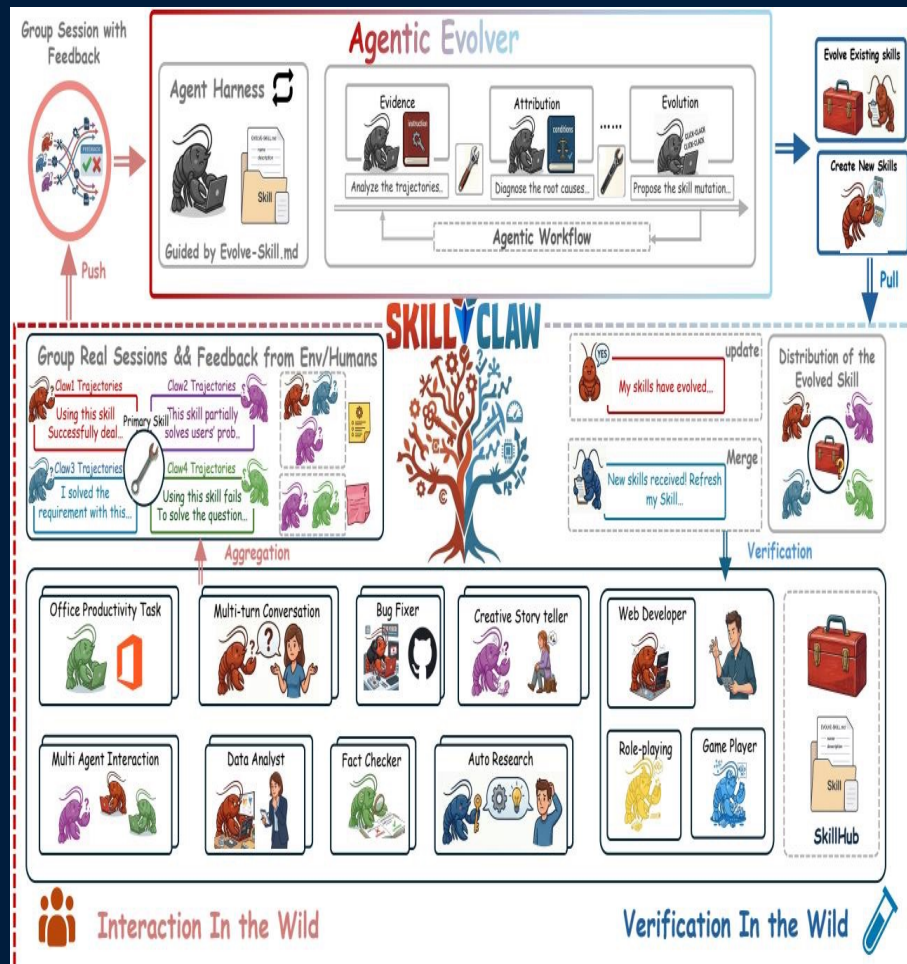


SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Let Skills Evolve Collectively with Agentic Evolver



Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

⚠ Static Skill Systems

- Skills manually installed from central hub
- No updates from runtime experience
- Failures & solutions not persisted
- Same problems rediscovered repeatedly
- Each user starts fresh — no system learning

✗ Existing Approach Gaps

- Memory methods: tied to specific instances
- Hard to generalize past trajectories
- Skill libraries treated as static resources
- No bridge from experience to evolution
- Knowledge cannot accumulate system-wide

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

Agentic evolution driven by open-ended reasoning — not predefined rules



↻ Continuous closed-loop: skills improve with every user interaction cycle



Fig 1: SkillClaw architecture

Agentic Evolver: Evidence → Attribution → Evolution

Open-ended agent reasoning — not predefined update rules

1

EVIDENCE

Analyze cross-user interaction trajectories to identify behavioral patterns: where skills succeed and where they fail across diverse contexts

2

ATTRIBUTION

Diagnose root causes of failures and successes — tracing observed behaviors back to specific skill instructions or environmental conditions

3

EVOLUTION

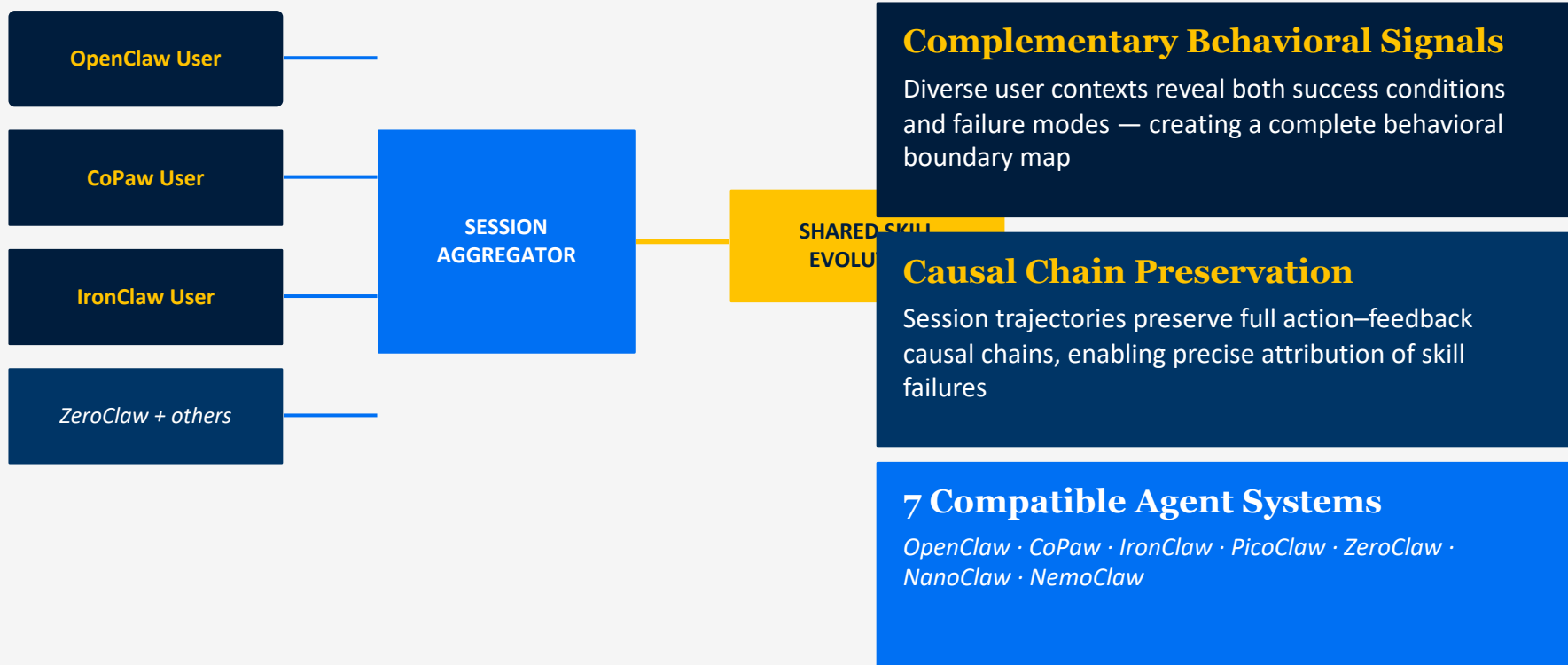
Propose skill mutations: refine existing skills or synthesize new ones. Verified before merging into the shared skill repository

Why Agentic Evolution?

- Open-ended reasoning adapts to novel failure modes
- No hand-coded rules to maintain or update
- Complementary signals from diverse user contexts
- Single user cannot separate generalizable improvements from idiosyncratic fixes
- Conservative verification prevents regressions

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Single users cannot separate generalizable improvements from idiosyncratic fixes



Case Study: Multimodal Creative Pipeline — 6 Days of Evolution Attempts

Conservative verification: only 1 of 6 candidates accepted over 6 days

1/6

Day	Candidate Skill	Outcome	Next-Day Action
1	validate-tmp-workspace-inputs	ACCEPTED ✓	Upgrade to new best pool
2	multimodal-input-validation-and-setup	REJECTED ✗	Keep current best pool
3	multimodal-creative-task-pipeline	REJECTED ✗	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	REJECTED ✗	Keep current best pool
5	creative pipeline + per-file fail-fast validation	REJECTED ✗	Keep current best pool
6	multimodal-creative-task-pipeline (cand.) — extended PDF-to-poster paths	REJECTED ✗	Not admitted to next cycle

Takeaway: High rejection rate reflects a conservative safety-first verification process that prevents skill regressions

Case Study: Git Push Auth-Fallback — Iterative Refinement Over 6 Days

3 of 6 candidates accepted — iterative refinement before reaching a plateau

3/6

Day	Candidate Skill / Change	Outcome	Next-Day Action
1	git-push-with-auth-fallback — safe fallback instead of blocking on push failure	ACCEPTED ✓	Add to Safety best pool
2	git-push-with-auth-fallback — unified patch/bundle filenames, reduced inconsistency	ACCEPTED ✓	Keep updated best pool
3	auth-alternative paths & secrets audit; fixed subshell pitfalls	ACCEPTED ✓	Keep current best pool
4	<i>(retest) — same-pool retest; validator confirmed current pool as best</i>	ACCEPTED ✓	Continue same best pool
5	Added push hang treated as auth failure — no improvement observed	REJECTED ✗	Keep current best pool
6	Added identity config & filename consistency — no improvement observed	REJECTED ✗	Not admitted to next cycle

Takeaway: Iterative refinement succeeds when evidence supports it — 3 accepted improvements before natural plateau

Three Properties That Distinguish SkillClaw

Implications for Continuously Improving Agent Systems

01

COLLECTIVE EVOLUTION

Individual interactions contribute to a shared, continuously improving skill ecosystem. No knowledge is wasted — every user session advances the collective.

02

FULLY AUTOMATIC

Skill evolution driven entirely by runtime interaction — no manual curation or explicit user intervention required at any stage.

03

AGENTIC EVOLUTION

Skill updates produced through open-ended reasoning over interaction evidence — not predefined rules. The evolver adapts to novel failure modes.

Implications

- Evaluated on WildClawBench with Qwen3-Max backbone — substantial improvements across tasks
- Cross-user aggregation is key: single users cannot generate sufficient signal for generalizable skill improvements